

Overview

This documentation describes the deployment of an Azure Application Gateway in front of Imaging v3.

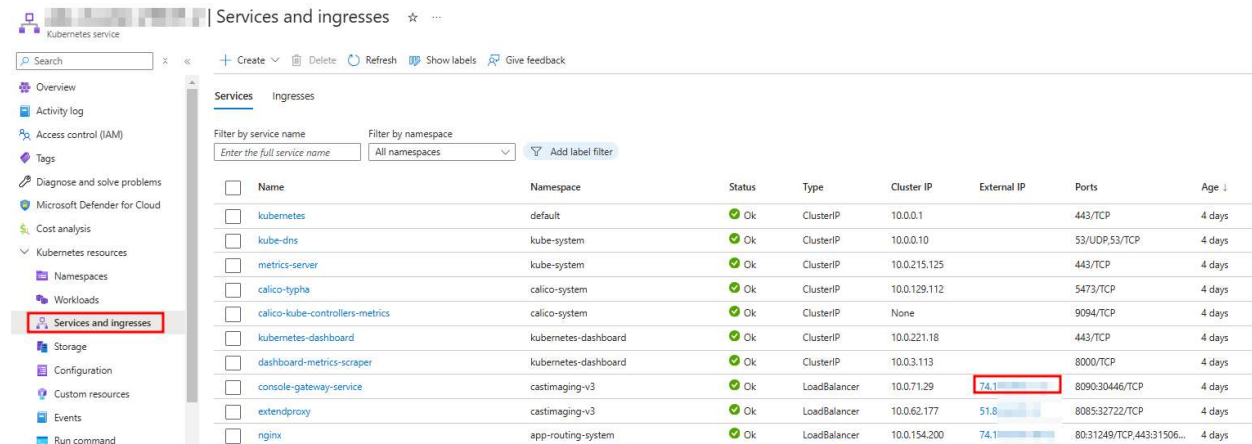
Prerequisites

- You own a valid certificate in the PFX format with the associated passphrase.
- You have the required permissions on your Azure tenant to create Application Gateways.
- You have the required permissions to create a DNS record on the desired DNS zone which will point to your Application Gateway listener IP address.

Procedure

Retrieve console-gateway-service external IP

Retrieve the console-gateway-service pod external IP.



Kubernetes service

Search

+ Create - Delete Refresh Show labels Give feedback

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Microsoft Defender for Cloud
Cost analysis
Kubernetes resources
Namespaces
Workloads
Services and ingresses
Storage
Configuration
Custom resources
Events
Run command

Services Ingresses

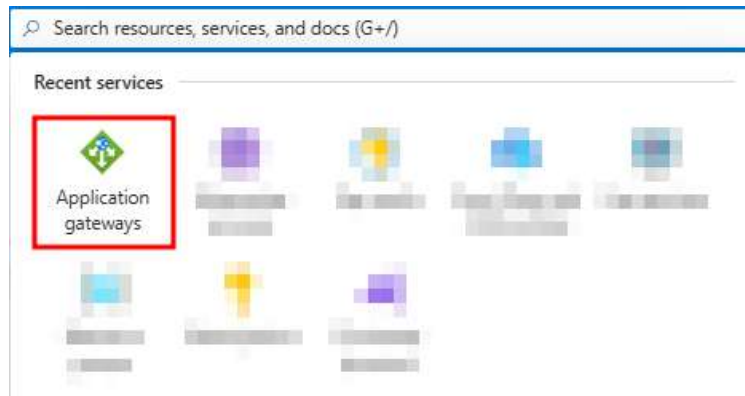
Filter by service name Filter by namespace

Enter the full service name All namespaces Add label filter

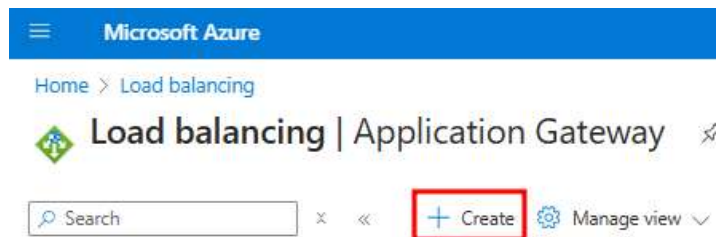
☐	Name	Namespace	Status	Type	Cluster IP	External IP	Ports	Age
☐	kubernetes	default	Ok	ClusterIP	10.0.0.1		443/TCP	4 days
☐	kube-dns	kube-system	Ok	ClusterIP	10.0.0.10		53/UDP,53/TCP	4 days
☐	metrics-server	kube-system	Ok	ClusterIP	10.0.215.125		443/TCP	4 days
☐	calico-typha	calico-system	Ok	ClusterIP	10.0.129.112		5473/TCP	4 days
☐	calico-kube-controllers-metrics	calico-system	Ok	ClusterIP	None		9094/TCP	4 days
☐	kubernetes-dashboard	kubernetes-dashboard	Ok	ClusterIP	10.0.221.18		443/TCP	4 days
☐	dashboard-metrics-scraper	kubernetes-dashboard	Ok	ClusterIP	10.0.3.113		8000/TCP	4 days
☐	console-gateway-service	castimaging-v3	Ok	LoadBalancer	10.0.71.29	74.111.111.111	8090:30446/TCP	4 days
☐	extendproxy	castimaging-v3	Ok	LoadBalancer	10.0.62.177	51.8.111.111	8085:32722/TCP	4 days
☐	nginx	app-routing-system	Ok	LoadBalancer	10.0.154.200	74.111.111.111	80:31249/TCP,443:31506/TCP	4 days

Create the Application Gateway

Search for the Azure Application Gateways service.



Create a new one.



Set the usual settings as desired. You will need to use a subnet dedicated to Application Gateways. Only use IPv4 IP address types as IPv6 is not yet supported on Imaging.

Microsoft Azure

Home > Load balancing | Application Gateway >

Create application gateway

1 Basics

2 Frontends

3 Backends

4 Configuration

5 Tags

6 Review + create

An application gateway is a web traffic load balancer that enables you to manage traffic to your web application. [Learn about creating application gateway](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group *

Create new

Instance details

Application gateway name *

Region *

Tier

Availability zone *

IP address type

HTTP2

my-app-gw

East US

Basic

Zones 1, 2, 3

☒ IPv4 only ☐ Dual stack (IPv4 & IPv6)

☐ Disabled ☒ Enabled

Configure virtual network

Virtual network *

Subnet *

Create new

Manage subnet configuration

vnet-app-gw-

vnet-app-gw- subnet1 (10.0.1.0/24)

Previous

Next : Frontends >

Configure a new IP address or use an existing one.

 Microsoft Azure

Home > Load balancing | Application Gateway >

Create application gateway ...

✓ Basics

2 Frontends

3 Backends

4 Configuration

5 Tags

6 Review + create

Traffic enters the application gateway via its frontend IP address(es). An application gateway can use a public IP address, private IP address, or one of each type. [↗](#)

Frontend IP address type ⓘ ☒ Public ☐ Private ☐ Both

Public IPv4 address * 

[Add new](#)

Create a new backend pool. Use the console-gateway-service external IP you retrieved above.

Add a backend pool.

 Basic SKU supports a maximum of 5 backend pool targets

A backend pool is a collection of resources to which your application gateway can send traffic. A backend pool can contain virtual machines, virtual machines scale sets, IP addresses, domain names, or an App Service.

Name * 

Add backend pool without targets ☐ Yes ☒ No

Backend targets

1 item

Target type	Target
IP address or FQDN	10.0.0.1   
IP address or FQDN	

Configure the listener. Choose the “Upload a certificate” option and upload a certificate which will be valid for the FQDN you plan to use for the Application Gateway.

Home > Load balancing | Application Gateway >
Create application gateway ...

✓ Basics ✓ Frontends ✓ Backends **Configuration** ③ Tags ④ Review + create

Create routing rules that link your frontend(s) and backend(s). You can also add more backend pools, add a second frontend IP configuration if you haven't already, or edit previous configurations. ⓘ

Frontends
+ Add a frontend IP
Public (new) my-app-gw-pub-ip

Routing rules
+ Add a routing rule

Public (new) my-app-gw-pub-ip

Previous Next: Tags >

Add a routing rule

Basic SKU supports a maximum of 5 routing rules

Configure a routing rule to send traffic from a given frontend IP address to one or more backend targets. A routing rule must contain a listener and at least one backend target.

Rule name * my-app-gw-routing-rule ✓

Priority * 100 ✓

* Listener * Backend targets

A listener "listens" on a specified port and IP address for traffic that uses a specified protocol. If the listener criteria are met, the application gateway will apply this routing rule. ⓘ

Listener name * my-app-gw-listener ✓

Frontend IP * Public IPv4 ✓

Protocol HTTP HTTPS ✓

Port * 443 ✓

Https Settings

Choose a certificate ☒ Upload a certificate ☐ Choose a certificate from Key Vault

Cert name * my-app-gw-certificate ✓

PFX certificate file * pfx ✓

Password * ✓

Listener type Basic Multi site

Custom error pages

Show customized error pages for different response codes generated by Application Gateway. This section lets you configure Listener-specific error pages. [Learn more](#) ⓘ

Please verify that the url(s) being added here is reachable from your application gateway using the [connection troubleshooting](#) tool to prevent any deployment error.

Bad Gateway - 502 Enter HTML file URL

Forbidden - 403 Enter HTML file URL

Add Cancel

In case the Application Gateway is shared with other applications, use this listener setting:

Listener type ⓘ

☐ Basic ☒ Multi site

Host type ⓘ

Single Multiple/Wildcard

Host name * ⓘ

abc.xxxxxxxxxx.com ✓

Configure the backend target with the following backend setting.

Add Backend setting



Basic SKU supports a maximum of 5 routing rules

[← Discard changes and go back to routing rules](#)

Backend settings name *	my-app-gw-backend-setting	✓
Backend protocol	☒ HTTP ☐ HTTPS	
Backend port *	8090	✓

Additional settings

Cookie-based affinity ⓘ	☐ Enable ☒ Disable
Connection draining ⓘ	☐ Enable ☒ Disable
Request time-out (seconds) * ⓘ	20
Override backend path ⓘ	

Host name

By default, the Application Gateway sends the same HTTP host header to the backend as it receives from the client. If your backend application/service requires a specific host value, you can override it using this setting.

	☐ Yes ☒ No
Override with new host name	☐ Yes ☒ No
Create custom probes	

Add a routing rule



i Basic SKU supports a maximum of 5 routing rules

Configure a routing rule to send traffic from a given frontend IP address to one or more backend targets. A routing rule must contain a listener and at least one backend target.

Rule name * ✓

Priority * ⓘ ✓

* Listener * Backend targets

Choose a backend pool to which this routing rule will send traffic. You will also need to specify a set of Backend settings that define the behavior of the routing rule. ⓘ

Target type ☒ Backend pool ☐ Redirection
 ▼

Backend target * ⓘ [Add new](#)

▼

Backend settings * ⓘ [Add new](#)

Path-based routing

You can route traffic from this rule's listener to different backend targets based on the URL path of the request. You can also apply a different set of Backend settings based on the URL path. ⓘ

Path based rules

Path	Target name	Backend setting name	Backend pool
No additional targets to display			

Review the settings and click on the create button.

Microsoft Azure

Home > Load balancing | Application Gateway >

Create application gateway

Validation passed

✓ Basics

✓ Frontends

✓ Backends

✓ Configuration

✓ Tags

6 Review + create

Basics

Subscription	sub_dpt
Resource group	rg_
Name	my-app-gw
Region	East US
Tier	Basic
Instance count	1
Availability zone	Zones 1, 2, 3
HTTP2	Enabled
Virtual network	vnet-app-gw-
Subnet	vnet-app-gw- subnet1
Subnet address space	

Frontends

Public IPv4 address name	my-app-gw-pub-ip
SKU	Standard
Assignment	Static
Availability zone	ZoneRedundant

Tags

None

Create

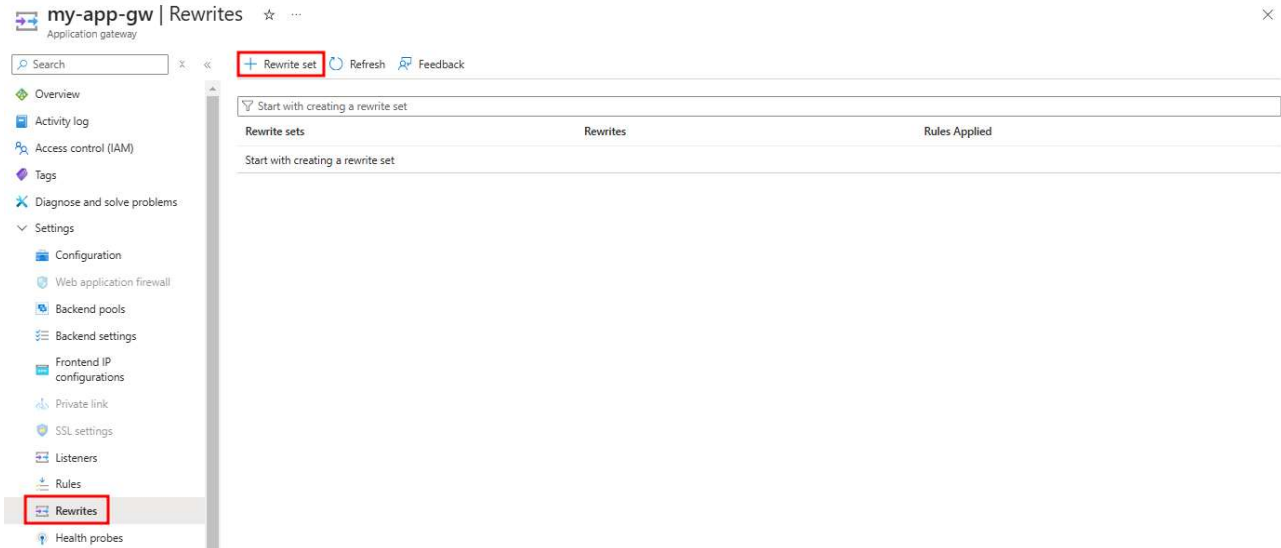
Previous

Next

Download a template for automation

Wait for your Application Gateway to be deployed, then create a new Rewrite set:

⇒ **Rewrites are only required if you are planning to enable SAML authentication in Imaging.**



Home > Load balancing | Application Gateway > my-app-gw | Rewrites >

Create rewrite set

1 Name and Association 2 Rewrite rule configuration

To rewrite HTTP(S) headers, you need to create rewrite sets and associate them with routing rules. On this tab, you can provide the name to the rewrite set and associate it with the routing rules in your application gateway. On the next tab, you can configure the rewrite set by adding one or more rewrite rules to it. [Learn more about rewrite sets.](#)

Name *

Associated routing rules

Select the routing rules to associate to this rewrite set. You can't select routing rules that already have an associated rewrite set.

Routing rules Paths	Type
☒ my-app-gw-routing-rule	Basic rule

Create the following rewrite rules:

[+ Add rewrite rule](#)

Rewrite rule name	Rule sequence	Conditions	Action				
			Rewrite type	Action type	Header name	Common header	Header value
cookie-rewrite	100	Condition#1: Type of variable to check: HTTP header Header type: Response Header Header name: Common header Common header: Set-Cookie Header value pattern to match: * Operator: equal (=) Case-sensitive: No Condition#2: Type of variable to check: HTTP header Header type: Response Header Header name: Common header Common header: Location Header value pattern to match: (.*)/auth/admin/(.*) Operator: not equal (!=) Case-sensitive: No	Response Header	Set	Common header	Set-Cookie	Header value pattern to match: * Operator: equal (=) Case-sensitive: No Set Header value to: {http_resp_Set-Cookie}; SameSite=None; Secure
XForwardedHost	100		Request Header	Set	Common header	X-Forwarded-Host	<your public Application Gateway FQDN> E.g. imaging.mydomain.com
XForwardedProtocol	100		Request Header	Set	Common header	X-Forwarded-Proto	https
XScheme	100		Request Header	Set	Custom header	X-Scheme	https
XForwardedScheme	100		Request Header	Set	Custom header	X-Forwarded-Scheme	https
XRealIP	100		Request Header	Delete	Custom header	X-Real-IP	
XForwardedFor	100		Request Header	Delete	Custom header	X-Forwarded-For	

Finally, you should see this set of rules:

Rewrite rules (Rule sequence)

cookie-rewrite (100)

XForwardedHost (100)

XForwardedProtocol (100)

XScheme (100)

XForwardedScheme (100)

XRealIP (100)

XForwardedFor (100)

cookie-rewrite rule details:

If

Type of variable to check * ⓘ
Header type *
Header name *
Common header *
Header value pattern to match ⓘ
Operator *
Case-sensitive * ⓘ

HTTP header
Response Header
☒ Common header
☐ Custom header
Set-Cookie
.*
equal (=)
☐ Yes ☒ No

OKCancel

And If

Type of variable to check * ⓘ
Header type *
Header name *
Common header *
Header value pattern to match ⓘ
Operator *
Case-sensitive * ⓘ

HTTP header
Response Header
☒ Common header
☐ Custom header
Location
(.*)/auth/admin/(.*)
not equal (!=)
☐ Yes ☒ No

OKCancel

Then

Rewrite type * ⓘ
Action type * ⓘ
Header name * ⓘ
Common header *
Header value pattern to match ⓘ
Operator
Case-sensitive ⓘ
Set Header value to * ⓘ

Response Header
Set
☒ Common header
☐ Custom header
Set-Cookie
.*
equal (=)
☐ Yes ☒ No
(http_resp_Set-Cookie); SameSite=None;...

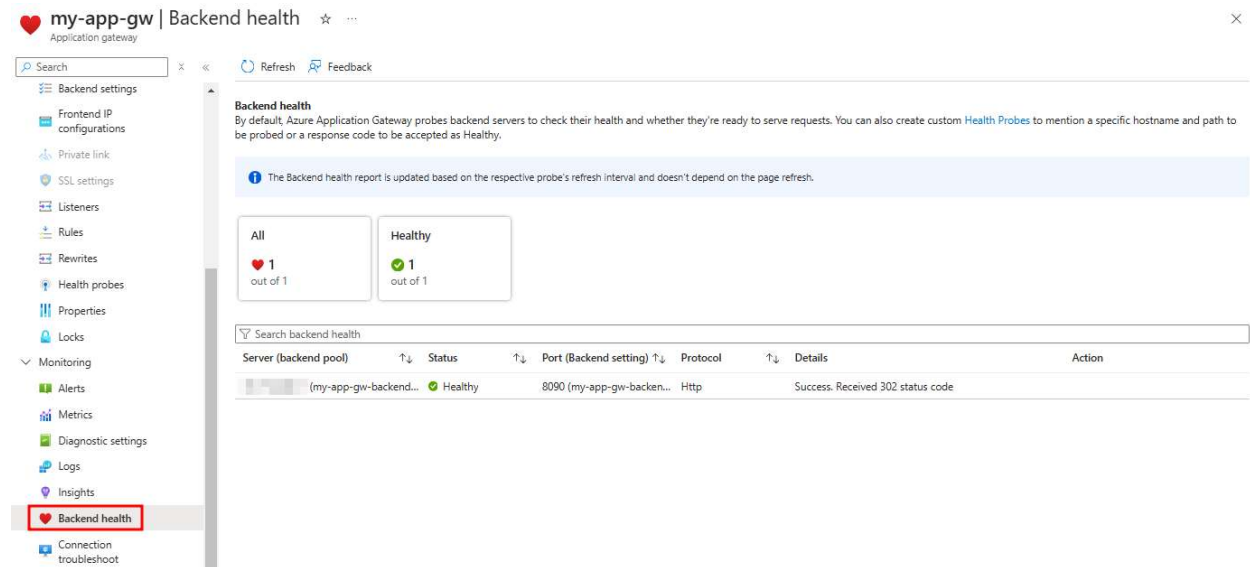
OKCancel

Create the Application Gateway DNS record

Make sure that the FQDN you plan to use for the Imaging deployment points to the Application Gateway public IP.

Resolve issues

If you encounter issues when connecting through the Application Gateway, carefully review all the procedure steps. Also confirm that the Application Gateway backend is healthy. It should return a 302 as below.



The screenshot shows the 'Backend health' page for an Application Gateway named 'my-app-gw'. The left sidebar contains a navigation menu with items like 'Backend settings', 'Frontend IP configurations', 'Private link', 'SSL settings', 'Listeners', 'Rules', 'Rewrites', 'Health probes', 'Properties', 'Locks', 'Monitoring', 'Alerts', 'Metrics', 'Diagnostic settings', 'Logs', 'Insights', 'Backend health' (highlighted with a red box), and 'Connection troubleshoot'. The main content area has a title 'my-app-gw | Backend health' and a subtitle 'Application gateway'. Below the title is a search bar and a 'Refresh' button. The 'Backend health' section explains that Azure Application Gateway probes backend servers to check their health. A status summary shows 'All 1 out of 1' and 'Healthy 1 out of 1'. A table titled 'Search backend health' displays the following data:

Server (backend pool)	Status	Port (Backend setting)	Protocol	Details	Action
(my-app-gw-backend...	Healthy	8090 (my-app-gw-backen...	Http	Success: Received 302 status code	